

Master: O25_RBS_I_DTS_DL**Subject: 03 - Artificial Intelligence, Machine Learning & Deep Learning (O25_RBS_I_DTS_DL)****Professor: Carmine Buono**

This assignment delves into the exciting world of Machine Learning (ML), Natural Language Processing (NLP) and Deep Learning. This assignment is structured into four levels of increasing complexity.

- Answer all questions clearly and concisely
- There is no need to write code
- Focus on understanding, reasoning, and critical thinking
- There is no single correct answer for open questions

LEVEL 1 – MULTIPLE CHOICE QUESTIONS (18 points)

Choose the correct answer.

1. What is Machine Learning?
 - a) A way to store data
 - b) A method to manually program decisions
 - c) A system that learns patterns from data
 - d) A database technology
2. What is supervised learning?
 - a) Learning without data
 - b) Learning with labeled data
 - c) Learning only images
 - d) Learning without output
3. Which of the following is a classification problem?
 - a) Predicting house price
 - b) Predicting temperature
 - c) Predicting if a customer will churn
 - d) Measuring distance
4. What does SVM stand for?
 - a) Statistical Variable Model
 - b) Support Vector Machine
 - c) System Vector Method
 - d) Simple Value Model
5. What is the goal of K-Means?
 - a) Predict values
 - b) Classify data
 - c) Group similar data points
 - d) Reduce noise
6. What is a neural network?
 - a) A database

- b) A set of rules
 - c) A model inspired by the human brain
 - d) A visualization tool
7. What is an activation function used for?
- a) Storing data
 - b) Adding randomness
 - c) Introducing non-linearity
 - d) Cleaning data
8. What is overfitting?
- a) Model too simple
 - b) Model too complex and memorizes data
 - c) Model too fast
 - d) Model too small
9. What is clustering?
- a) Predicting labels
 - b) Grouping similar observations
 - c) Removing data
 - d) Scaling data

LEVEL 2 – OPEN QUESTIONS (4 points)

Answer briefly.

1. Explain the difference between supervised and unsupervised learning.
2. Why is model interpretability important in Machine Learning?

LEVEL 3 – CONCEPTUAL QUESTIONS (4 points)

1. Explain in simple terms how a neural network learns from data.
2. When would you use K-Means instead of a classification model?

LEVEL 4 – BUSINESS CASE & CONCEPTUAL DESIGN (NO CODING) (4 points)

Design a Machine Learning Solution

Scenario:

You are working for a healthcare organization that wants to develop a system to support cancer diagnosis using patient data.

The dataset includes:

- demographic data
- lifestyle information
- medical history

The goal is to classify whether a patient is at risk.

Tasks:

Task 1 – Problem Understanding

- What type of machine learning problem is this?
- What is the target variable?
- Why is this problem important?

Task 2 – Data Understanding

- Which features do you think are most relevant?
- What potential data issues could exist (e.g. missing values, bias, imbalance)?

Task 3 – Data Preparation Strategy

Describe how you would prepare the data:

- missing values
- categorical variables
- scaling

Task 4 – Model Selection

- Which model would you choose (e.g. Logistic Regression, SVM, Neural Network)?
- Why is it appropriate?

Task 5 – Model Evaluation

- Which metrics would you use?
- Why is accuracy not enough in this context?

Task 6 – Interpretability

- Why is it important to explain the model's decisions?
- How could interpretability help in real-world applications?

Task 7 – Risks and Limitations

- What are the risks of using such a system?
- Consider bias, errors, and ethical implications

Task 8 – Business Impact

- How would this system support decision-making?
- Should it replace or support doctors? Why?

Grading Criteria

- **Level 1**
 - 18 points out of 30
- **Level 2**
 - 4 points out of 30
- **Level 3**
 - 4 points out of 30
- **Level 4**

- 4 points out of 30
- **Honors ("30 cum laude"):** 1 extra point to a work which is excellent in all its components.

IMPORTANT:

- **Be concise, NOT verbose; be schematic and conceptual, NOT discursive nor vague.**
- **When you draw up a technical document, a good practice is to use Courier New font for the code and the results**
- **Use bullet points to list deliverables and objectives.**
- **Assignments that are clearly generated using AI tools (e.g., generative AI systems) and do not reflect the student's own reasoning and understanding will not be evaluated. The purpose of this assignment is to assess your personal critical thinking and comprehension of the topics covered during the course. Students are therefore expected to submit original work based on their own analysis.**